Detection of the Top Anemic Diseases in Blood Smear Images Using Image Quantization Followed by Ensemble of Classifiers

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ABSTRACT
Anemia is a condition caused due to the deficiency of Red Blood Cells (RBCs) and hemoglobin in blood. It is an indication to a specific disorder in the human body. Different types of anemic diseases infect the shapes of Red Blood Cells in different ways and the infected cells form various geometric shapes, such as elongated ellipse, triangular shapes, cut circles, boundary interruption in ellipse or circle etc. Leveraging these shapes the type of anemia can easily be identified. We have used various boundary based shape descriptors like shape signatures and color profiles as features for the infected RBCs recognition. The algorithm is followed by preprocessing steps like color channel separation, segmentation through quantization, feature extraction and finally classification of Red Blood Cells and the diseases associated with them. We have achieved 92 % accuracy and the proposed method is cost effective and easy to use.

CCS Concepts
• **Applied computing → Imaging**

Keywords
Anemia, RBCs, descriptors, shapes, segmentation, classification, hemoglobin.

1. INTRODUCTION
Anemia is an abnormal condition of RBCs, where the RBCs change their shape due to some alterations like osmotic pressure, in the internal cell and this deformability plays a vital role in the detection of anemia, which include comprehensive review papers [1, 2].

In evaluating a patient with anemia, it is necessary to determine if the patient is clinically stable or unstable, RBC transfusion may be necessary in patients with symptomatic anemia i.e. angina, hypotension, shortness of breath palpitation, syncope, pre-syncope. Anemia is due to one of the processes - bleeding, bone marrow production problems, hemolysis, which is excessive Red Blood Cells Breakdown. Hospitalize or complex patients may have multi factorial anemia with the combination of the above processes.

Anemia and the diseases associated with it can be classified by RBCs size, e.g. microcytic anemia can be due to thalassemia, iron deficiency and anemia of chronic diseases. Macrocytic anemia can be caused due to B12 or Folate deficiency, alcohol use, chronic liver diseases such as methotrexate, reticulocytosis, aplastic anemia, myelodysplastic syndrome and hypothyroidism. Normocytic anemia may involve a combination of factors and is best evaluated using mechanistic approach. In cases of acute anemia, the blood smear must be reviewed to access for fragmentation of Red Blood Cells which may indicate possible life threatening conditions such as thrombotic thrombocytopenic purpura or the presence of blast cells which are immature White Blood Cells seen in acute leukemia.

The images of anemic peripheral blood smear containing cells which can be distinguished visually based on the following properties and that are of our interest in image processing:

Size: The sizes of each primary classes of RBC are microcytes having diameter of about 6 – 7.5 micrometer, Normocytes are 7.5-8 micrometer and that of macrocytes are 8 – 9 micrometers.as shown in Figure 1 (a,b,c respectively).

Hemoglobin distribution: The central pallor of each class varies as normochromic, hypochromic and hyperchromic

Shape: The shapes of cells vary due to disorders in the body as elliptic, spur, burr, tear drops, helmet etc. shown in Figure 1(k through t).

Inclusions: Some inclusions like peppen heimers, cabor ring, bluish stiplings and howel jolly shown in Figure 1(g to j) may be found in these cells due to disorders.

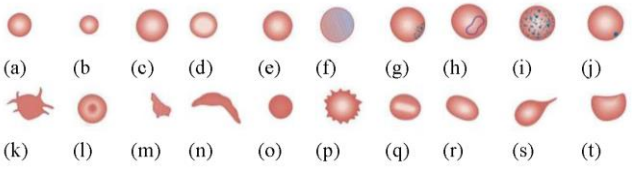


Figure 1.(a) microcyte (b) normocyte (c) macrocyte (d) hypochromic (e) normochromic (f) Polychromic (g) peppen heimers (h) cabor ring (i) bluish stiplings (j) howel jolly (k) Spurr (l) Target (m) Schistocyte (n) Sickle (o) Spherocyte (p) Burr (q) Stomatocyte (r) Elliptocyte (s) Teardrop (t) Helmet.

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2. RELATED WORK

Anemic diseases have been widely investigated by researchers all around the world through the analysis of blood smear images. Some of the most prominent works are presented here.

Segmentation of RBC and WBC is done by Sharif et al. [3] in blood images by filtering noise and enhancing image quality, formerly transformed the RGB color space to YcBcR, then adjusted its contrast and smoothen the image using median filter, then morphological operations like erosion, dilation, opening and closing were used, at this stage, WBC were extracted, using masking operation WBC and other small particles were subtracted and obtained RBCs, after obtaining RBCs watershed algorithm was used for separating overlapping cells. The peripheral blood smear images were processed by Mark et al. [4] for the detection of plasmodium by following the conversion of RGB to grayscale, then median filter for noise removal, then used morphological closing using disk shaped structuring element for the background extraction, then RBCs sizes were estimated, then using zack [5] thresholding foreground was extracted. For the cells determination in the clumped area they calculated its curvature suggested by [6]. For the detection of malarial parasites in the stained blood smear images, Bhowmik et al.[7] used the morphological area opening for removing small objects, then area of each component was calculated, and all the RBCs detected were displayed and the inclusion in each RBC were screened and sobel edge detection algorithm was applied in order to find malarial parasites. Circlet transform is used by Omid et al. [8] for the segmentation of RBCs after applying median and Gaussian filter of window 3x3 with a standard deviation 10 for smoothing the image, the color space conversion from RGB to Lab along with K Mean clustering algorithm was done to extract WBCs and RBCs. Poikilocytes were segmented by Tyagi et al. [9] using Artificial Neural Network subsequently by enhancing the contrast of the gray scale image, finding edges using Canny Edge detection technique, morphological filling for filling the small holes in the cells, then removed boundary and touching cells and finally extracted some features for putting them into ANN Classifier. Another closely related work is done by Mahsa et al. [10] for the detection of three types of abnormal RBCs i.e. Dacrocyte, Schistocyte and Elliptocyte cells by following the histogram equalization, binarization with four combined techniques Sovala, Niblack, histogram and Otsu thresholding, after segmentation the features were collected by them and were tested by three types of classifiers - neural network, KNN and SVM. Similarly, watershed technique used by Hari et al. [11] for the separation of connected cells and some geometric features for the classification of WBCs, RBCs and platelets. In the work of Jullend et al. [12] for the detection of plasmodium parasites in blood cells images, they used Median filter (3x3) and used Otsu thresholding method for the binarization of whole image, segmented the RBCs from the rest of the image and in another step, they used another thresholding step for the detection of plasmodium in the nucleus of RBCs. Hemolytic Anemia is a type of RBC disorder and Pranati et al. [13] worked for its automatic diagnosis, where they used sobel operator after binarizing the image for the detection of cells edges and found their roundness, and thresholded the roundness value for the identification of bite cells, an indication to the Haemolytic anemia. Mazalan et al. [14] after getting a binary image, used circular Hough Transform and extracted the RBCs from the rest of image.

In our work, the consistency against the variant image factors like luminosity, contrasts and brightness etc. was focused, which was achieved through our quantization technique up to a maximum level. Moreover, an extension has been made to the existing work in the feature extraction area. We have explored more features to make the automated identification of RBCs more accurate. For the purpose, we used simple shape descriptors, chain codes, difference codes, Fourier Shape Descriptors, Centroid Distance Functions, Contour Curvatures, Area function, shape matrix, shape context, skeletons of shapes, Shock graphs, moment invariant, geometric moments, shape descriptors as moments etc.

3. PROPOSED METHODOLOGY

Our proposed plan of work is consisted of the main steps given in Figure 2. It principally focuses on the accurate extraction of the shapes given in the Figure 1.

Input images have been obtained from a local Provincial Research Hospital and the desired objects were marked by expert hematologists.

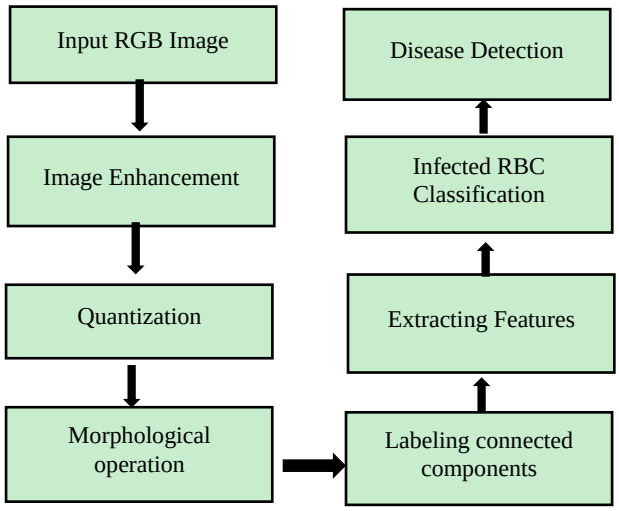


Figure 2. Work Flow of the Algorithm.

3.1 Image Enhancement

We used the Contrast-Limited Adaptive Histogram Equalization (CLAHE) technique [15], which enhances images using a small square of region rather than entire image. The result of each square is then combined with the help of bilinear interpolation to remove artificially induced boundaries. The contrast of homogeneous area may be limited to restrict the noise amplification if present in an image.

3.2 Quantization

Quantization is the process of reduction of color levels in an image, we have used this process for reducing the color complexities in an image, the process we used is given by equ. (1), (2) and (3);

$$Or = Fr * \text{round}\left(\frac{\text{Ir}}{\text{Fr}}\right) \quad (1)$$

$$Og = Fg * \text{round}\left(\frac{\text{Ig}}{\text{Fg}}\right) \quad (2)$$

$$Ob = Fb * \text{round}\left(\frac{\text{Ib}}{\text{Fb}}\right) \quad (3)$$

Where **r,g,b** are input channels of RGB image respectively. F is image factor, may be same for all 3 channels or varied ranging between 0 and 255. The quantized image was obtained by

dividing the image into intensity levels then normalizing it using same factor F. This process was repeated for all the three channels.

At end of this step we'll get image with only WBCs. The result is shown in Figure 3(b).

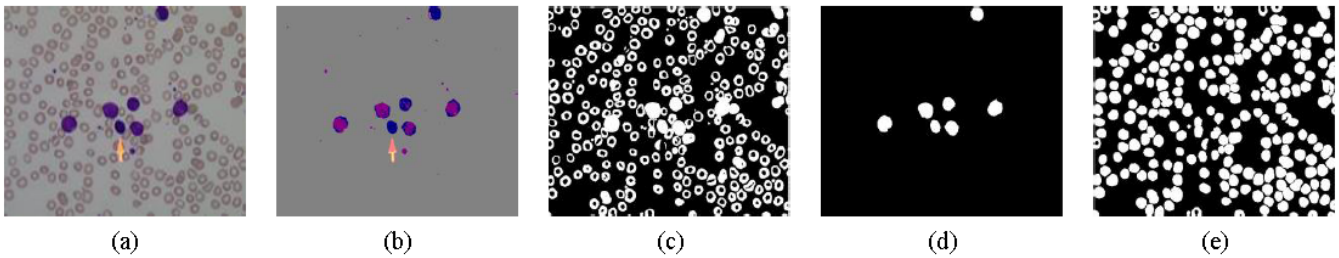


Figure 3. a) Original Blood Smear b) Quantized Image c) Binary Original Image d) Binary Quantized Image (WBCs)b e) XOR of quantized binary and original (RBCs).

3.3 Binarization

Binarization is an important step for converting an image into binary in order to segment it, we used a dynamic and automatic thresholding technique [16] called Otsu demonstrated by the following equations.

$$\sigma^2_1(t) = \sum^{t} [i - \mu_1(t)^2] \frac{P(i)}{\omega\ (t)} \quad (4)$$

$$\sigma^2_2(t) = \sum^{t} [i - \mu_2(t)^2] \frac{P(i)}{\omega\ (t)} \quad (5)$$

Where equ (4) represent threshold for the background pixels

3.4 Segmentation

It is easy for an algorithm to accurately segment an image if binarization step is done correctly, we require two segmentations, one for WBCs and other for RBCs in the following way:

3.4.1 WBC Segmentation

Right after quantization, we have left bind only WBCs, which is the simplest form of the blood smear image. The binarized image was obtained by thresholding this version of image with Otsu method, given by equ. 4 and 5. The result is shown in Figure 3(d).

3.4.2 RBC Segmentation

In order to get RBCs after quantization we took XOR of the binarized original and binarized quantized image, at this stage the RBCs obtained also include extra artifacts, which were removed by applying some morphological operations like opening, closing and dilation, the result can be seen in Figure 3(e).

3.5 Labeling the Objects

After that we got a binary image, another important step is to label the connected pixels (objects) with 4-connected neighborhood value in the image to associate a specific object with a vector of features.

3.6 Hemoglobin Concentration

After we have extracted the RBCs from the rest of image, it's time to find the measurement of hemoglobin inside each cell, it was found by measuring the inner blob of each cell, the pixels measuring more than one third of whole pixels of a cell is counted as hypochromic, while the one more than one third is categorized as hyperchromic and the rest are normochromic. This property of RBC is being used as a primary feature for anemia classification as shown in Figure 4.

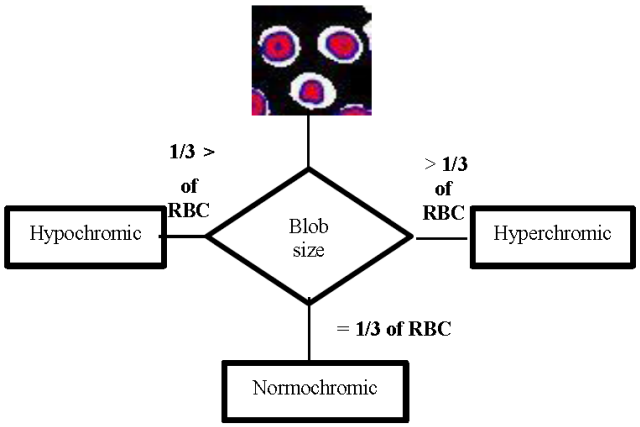


Figure 4. Classification of RBCs per Hemoglobin Concentration.

3.7 Features Extraction

Feature extraction is an important step in almost every type of shape recognition process. This work includes extraction of global and structural based features of both region and contour of shapes. The features we have extracted are described below:

3.7.1 Aspect Ratio

This feature is very useful for differentiating the circular or noncircular or needle like shapes. The value of Aspect Ratio ranges from 1 to 0. The value closer to 1 will be more elongated (like sickle cells in our case) than the one closer to 0 (like normocytes, macrocytes and microcytes).

$$Aspect\ Ratio = \frac{S(Height)}{S(Width)} \quad (6)$$

3.7.2 Convexity

Convexity is the perimeter of convex hull divided by perimeter of the shape and is given by:

$$Convexity = \frac{P(CHull)}{P} \quad (7)$$

Where P (CHull) and P represents perimeter of Convex hull of a shape and perimeter of shape respectively.

3.7.3 Solidity

It is the degree of convexity or concavity of a shape and is calculated as:

$$Solidity = \frac{A(s)}{CHull} \quad (8)$$

3.7.4 Circularity

Circularity is a dimensionless value which means how much an object is like a circle, depending upon the smoothness of perimeter of a shape., given by:

$$Circularity = \sqrt{\frac{4\pi A}{p^2}} \quad (9)$$

Where A is Area and P is perimeter.

3.7.5 Eccentricity

The distribution of pixel region around a centroid is determined by eccentricity. The eccentricity of squared and circular shapes close to 1, whereas the irregular will be closer to 0. It may be calculated with equation 6.

$$Eccentricity = \frac{Major\ Axis\ Length}{Minor\ Axis\ Length} \quad (10)$$

Table 1. Geometric features of some common RBCs

RBC Type	Aspect Ratio	Circularity	Convexity	Solidity
Normocytes	1	1	1	1
Tear Drop	0.10	0.49	1	1
Elliptocytes	0.22	0.69	1	1
Microcytic Elliptocytes	0.23	0.71	1	1
Target Cell	0.11	0.49	0.99	0.85
Sickle Cell	0.20	0.14	0.89	0.10
Overlaped Cells	0.50	0.75	0.87	0.88
Fragment Cell	0.94	0.86	0.94	0.91

3.7.6 Extent

Extent is a degree to which a shape fills a minimum bounding rectangle. It is the area of shape divided by area of bounding box.

3.7.7 Shape Signature

Shape signature portrays the perceptual features of the shape and often used as preprocessing step for the other feature extraction algorithms like the Fourier descriptors. It can be derived from coordinates of a shape boundary [17]. Some of the shapes given in Figure 5 may be distinguished by taking the histogram of their boundary points from their centroid as shown in Figure 6.The Shape signatures used here are Centroid Distance Function, Chord Length Distance, Farthest Point Distance, Angular Function and Triangular Centroid Area.

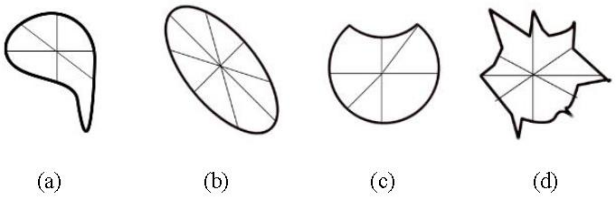
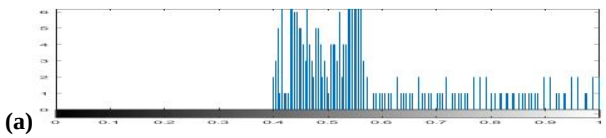


Figure 5. (a) Teardrop (b) Ovalocyte (c) Bite cell (d) Acanthocyte.



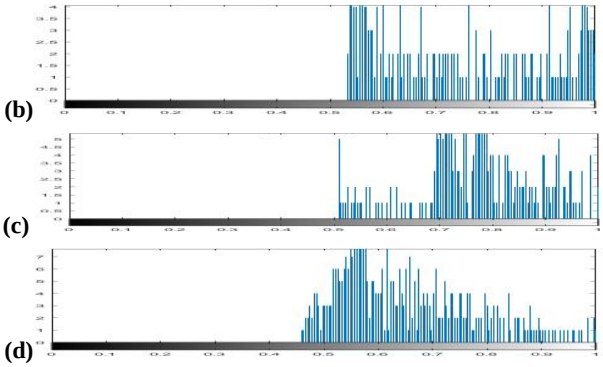


Figure 6. Histograms of boundary points from centroids. (a) Teardrop (b) Ovalocyte (c) Bite cell (d) Acanthocyte.

3.8 Classification

One of the modern techniques now a day is to combine two or more best available classifiers [18] in order to improve the prediction performance. It is possible in several ways. i.e. bagging, boosting, stacking and voting. In our work, we have selected the voting method of combining the results of classifiers due to frequently giving good results. We have selected two classifiers statically Multilayer Perceptual Model (MLP) and Random forest whose results were analyzed individually and then were ensembled through voting technique.

3.8.1 Multilayer Perceptual Model

One of the classifiers selected to be used in our algorithm is Multilayer Perceptual Model [19], which uses backpropagation to classify attributes. The network can also be monitored and modified during training time. The input (set of RBCs features) from the input layer is transmitted to the hidden layer containing nodes which is processed by applying a sigmoid function in case of non-numeric data and threshold values in case of numeric data, the output is stored in the output layer as classified data.

3.8.2 Random Forest

Random forest [20] itself is an ensemble classifier comprising of many decision trees and outputs the class that is the mean of the class's output by individual trees as demonstrated in figure 7. It is faster and accurate to predict the favorite class.

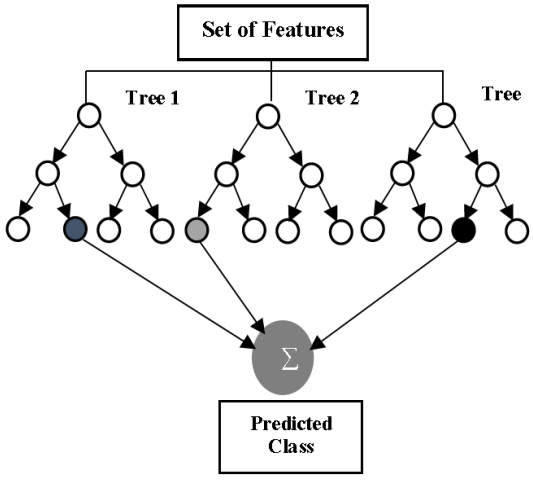


Figure 7. Prediction through Random Forest.

3.8.3 Voting

Ensemble learning is used to evaluate and combine the prediction of multiple individual classifiers [21], which in return attain better performance than an individual one. Since that different base classifiers result different contribution to the final classification result. The method we selected is to assign greater weights to the classifiers with better performance and proposes a weighted voting approach , can be seen in the Figure 8.

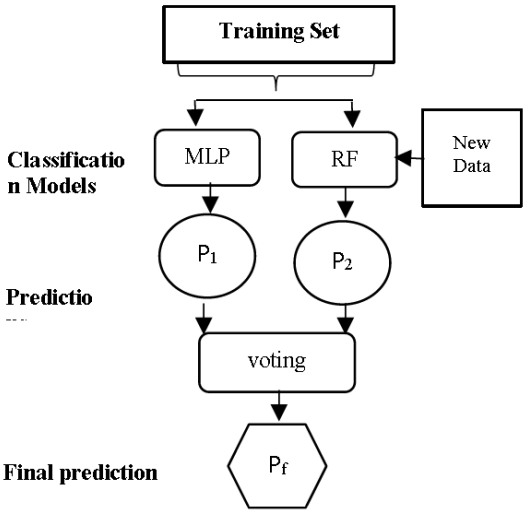


Figure 8. Final prediction through voting.

4. RESULTS

In our work, we have created three datasets at a local hospital based on different RBCs' shapes present in the images and image condition like luminosity, contrast, noise etc. The proposed technique was applied randomly on each image; the results were so promising as, one of the good features of our technique is the

reduction of the issues due to image condition variations as compared to other techniques, with the help of quantization. The proposed technique was tested and compared to other existing technique explained in section 2 of this paper and found better than them.

Table 2. Result of one of the train dataset

No. of Instances	Correctly Classified instances	Root Mena Absolute Error	Relative Absolute Error	Kappa Statistics
65	65	0.0747	18.244%	**1**

Table 3. Result of one of the test dataset.

No. of Instances	Correctly Classified instances	Root Mena Absolute Error	Relative Absolute Error	Kappa Statistics
46	42	0.1154	29.916 %	0.9045

The necessary information for a disease to be classified can be induced through frequency of different RBC shapes in blood smear as in Iron Deficiency Anemia, hypochromic microcytes are 54-62%, Elliptocytes are 12-15%, Normocytes are 9-11% target and tear drop are7-9%, similarly in Hemolytic Anemia the Hypochromic Macrocytes are 35%, spherocytic are 29 to 39 % Ovalocytes are 9 to 15% of the l abnormal cells, shown graphically in Figure 9 and 10. Similarly each anemic disease has its own frequency of specific cells found in a peripheral blood smear images, which will be used for the classification of anemic diseases.

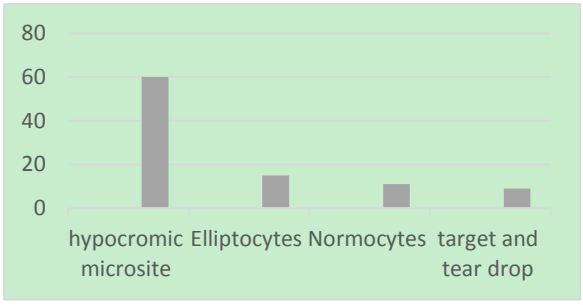


Figure 9. An Area Graph Showing the frequency of Cells in Iron Deficiency Anemia.

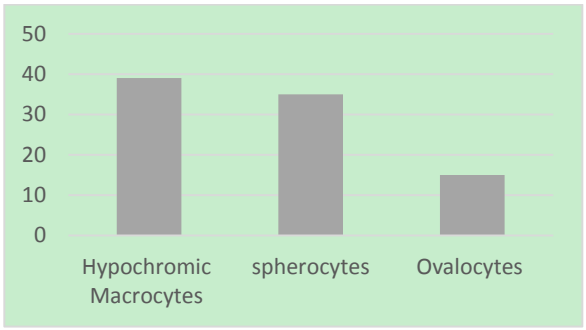


Figure 10. An Area Graph Showing the frequency of Cells in Hemolytic Anemia.

5. CONCLUSION

This paper aims to introduce and distinguish the top classes of anemia by segmenting Red Blood Cells in a peripheral blood smear using image quantization followed by a dynamic thresholding method, Otsu. Combinedly these two can cope with varied and uneven luminosity effects in images which suffer the accuracy of ultimate result of segmentation. We have used 8 selected features for the classification of different shapes of Red Blood Cells using ensemble of 2 reputed classifiers (Multi-Layer Perceptual model and Random Forest) using simple majority technique. The above results shown in table 2 and 3 confirm its accuracy and precision. This accuracy and precision is persistent for almost every sort of luminosity conditions, but under worst cases, if the result is not corresponding to the actual results of hematologists the factor for quantization may be changed manually. The work done so far may easily quantify the RBCs abnormality. In future, this work may be leaded to create a hematology ruler for anemic diseases identification.

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